PiCal — Pipe Weight Report

Bare-pipe mass (geometry, not a pressure-design rule)
m/L = pi t (D - t) rho / 1,000,000 [kg/m]
m = (m/L) L; ID = D - 2t
m_water = pi ID^2 L / 4,000 [kg], rho_water = 1000 kg/m^3

PROJECT	QA weight
ENGINEER	Device preview
DATE	06/10/2026

INPUTS

Material	Carbon steel
Nominal Pipe Size (NPS)	4 inch
Outside Diameter (OD)	114.3 mm
Wall Thickness (t)	6.02 mm
Density	7850 kg/m³
Pipe Length	6 m

OUTPUTS

UNIT WEIGHT
16.08 kg/m

TOTAL PIPE WEIGHT
96.45 kg

WATER FILL WEIGHT
49.28 kg

TOTAL (INCL. WATER)
145.73 kg

Nominal bare-pipe mass = pi * t * (D - t) * density / 1,000,000 (kg/m).
For carbon steel at 7850 kg/m3, this agrees with the B36.10 mass coefficient.
Water density, when selected: 1000 kg/m3. Excludes coatings, insulation and fittings.